

YUANBAI LI

AI Algorithm Engineer - Medical Multimodal Learning / Computer Vision

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EDUCATION

Nanyang Technological University (NTU), Lee Kong Chian School of Medicine *Artificial Intelligence in Medicine* *Aug 2026 - Jul 2027*

- **Core Coursework:** Machine Learning/Deep Learning for Healthcare AI; Natural Language Processing and Large Language Models in Healthcare AI; Medical Imaging/Multimodal Learning and Medical Domain Model Integration.

University of Nottingham Ningbo China, Faculty of Science and Engineering *Computer Science* *Sep 2022 - Jul 2026*

- **Core Coursework:** Machine Learning, Artificial Intelligence Methods, Algorithms and Data Structures, C++ Programming, Software Engineering, and Computer Vision.

SKILLS & PATENT

- **Language:** IELTS 6.5 and CET-6; strong English literature-reading and academic-writing skills, with fluent daily spoken communication.
- **Technical Skills:** Proficient in **Python/C++**; strong engineering programming, experimental analysis, teamwork and communication skills; disciplined coding style and documentation awareness.
- **Tools & Frameworks:** Familiar with **Linux/Ubuntu, Git and CMake**; proficient with **PyTorch, OpenCV and Alphalens**.
- **Machine Learning & Medical AI:** Knowledge of classical machine learning, unsupervised clustering and feature dimensionality reduction; hands-on experience in **quantitative factor mining**, **strategy backtesting** and **model evaluation**. Familiar with data processing and modeling workflows for medical multimodal models, with **end-to-end** R&D capability from feature extraction and model fusion to parameter optimization. Familiar with Transformer architectures; proficient in PyTorch-based development and experimental validation for image segmentation and survival prediction.
- **Publication:** H2-Surv: Hierarchical Hyperbolic Multimodal Representation Learning for survival Prediction. ------ CVPR2026 (Fourth author)
- **Patent:** Algorithm R&D for **"A High-Precision 3D Printing Slice Generation Method for Large-Scale Fine-Particle Fluidization Structures"**: designed a pixel-level grayscale-smoothing algorithm and batch slicing workflow for DLP photopolymerization 3D printing microstructures, addressing micron-scale pore blockage and structural collapse; applicable to research 3D printing of porous media and catalyst carriers.

INTERNSHIP EXPERIENCE

Kunming Bocha Technology Co., Ltd. *Quantitative Trading Intern* *Jun 2024 - Sep 2024*

- **Factor Research:** Analyzed fundamental and technical factors across **350+** trading instruments using Python and Alphalens; constructed clustering features from factor IC time series using **wavelent transform**s, **normalization** and **PCA**.
- **Factor Screening:** Compared clustering methods using **silhouette**, **Calinski-Harabasz** and **Davies-Bouldin** scores; combined clustering outputs with raw-factor correlations to automate factor grouping and screen for redundancy.
- **System Development:** Developed and optimized components of an automated trading system and Python strategy-backtesting modules, connecting market-data processing, factor construction, backtesting and signal generation to support strategy evaluation with multiple machine-learning models.

PROJECTS

Multimodal Survival Prediction Based on Foundation Models *Project leader* *Sep 2025 - May 2026*

- **Overview:** Integrated three heterogeneous data types - whole-slide pathology images, genomic data and pathology-report text - into one model to predict patient survival risk.
- **Key Work:**
  - Organized and aligned **TCGA** multimodal data by matching **WSI, genomic data, pathology-report text and survival labels**; cleaned, tokenized and truncated reports.
  - Extracted patch-level pathology-image features with **UNI**, encoded genomic data with **SNN** and encoded pathology-report text with **BioClinicalBERT**.
  - Introduced a gated text-refinement module using **a sigmoid gate** to adaptively control text-feature contribution to the final fused representation.
  - Extended the **MOTCat** pathology-gene fusion backbone into a tri-modal framework and trained it with a discrete-time survival model and NLL loss.
  - Completed baseline reproduction, model extension, training tuxsning, ablation experiments and visualization analysis.
- **Outcome:** Evaluated the tri-modal framework using five-fold cross-validation across five TCGA cancer cohorts, achieving the highest C-index among the evaluated on BRCA, UCEC and GBMLGG. Improved C-index on UCEC by **13.3%** relative to MOTCat.

Interactive Automatic Image Annotation Platform Development *Front-end work* *Nov 2024 - Apr 2025*

- **Overview:** An interactive medical-image annotation platform extended from ScribblePrompt. Users annotate images through points, scribbles and bounding boxes in a web interface, and the platform automatically generates segmentation masks. **Open-source: https://github.com/YYYhan/Interactive-Automatic-Image-Labeling-Platform-Development**
- **Key Work:** Built annotation tools and inference workflows with **Python, PyTorch, Gradio, OpenCV and Kornia**; encapsulated the **ScribblePrompt U-Net model** to process point, box, scribble and existing-mask inputs consistently and output results. Implemented image upload, brush scribbling, click/box prompts, prediction overlay, undo, clear and model-switching interactions. Supported multiple medical-image models; test cases covered retinal vessels, skin lesions, spine, teeth, whole-body CT and white blood cells.
- **Outcome:** Delivered an open-source interactive annotation platform with multiple prompt types and model switching; tested workflows across six medical-image domains, including retinal vessels, skin lesions and whole-body CT.